Response to Reviewers

We thank both reviewers for their thorough and constructive feedback on our manuscript "Tool Choice drastically Impacts CRISPR Spacer-Protospacer Detection." The comments have significantly improved the quality and impact of our work. In this revised version, we present a large overhaul of the project that addresses the concerns raised. We note that due to scope and format constraints, not all changes are implemented in the main text, and some are available in the jupyter notebooks in the project repository.

Briefly, following the reviewer suggestions we modified the analyses:

1. **New tools and updated versions:** We added new tools to the benchmark (sassy, x-mapper and indelfree.sh) published after the original manuscript submission (and the preprint version). We also updated the already included tools to their latest versions, or changed certain parameters based on reviewer suggestions. Notably, sassy was instrumental in testing the effects of using edit distance (allowing indels) versus hamming distance (substitutions only) up to larger mismatches/edits.

2. **More detailed overview of the tested tools:** We now provide a brief overview in the introduction section of the general logic and types of sequence alignment/search tools. We also updated table 1 to include some of these features. Briefly, we explain the difference between exhaustive and heuristic tools, the different types of alignments and distance metrics (hamming and edit) they focus on, and the computational resources (namely memory and CPU time) they require.

3. **Explicitly addressing resource usage:** Because we have added two exhaustive tools (sassy and indelfree_bruteforce) to the analysis (when possible, see note below about subsampling), we now include a section with the runtime/memory usage (vs distance threshold and approximate search space size), and we explicitly discuss the tradeoffs of heuristic vs exhaustive tools.

4. **Explicitly suggesting tool choice:** To clarify the new results and different scenarios where spacer-protospacer might be investigated, we have expanded the discussion and conclusion sections to provide explicit, scale-dependent tool recommendations that factor in dataset size, the availability of validation / experiment design, and the expected performance.

5. **Updated spacer dataset:** We replaced the "real-data" CRISPR spacer set with a more curated, up-to-date set from iPHoP (June 2025 release), combining spacers from arrays detected in reference genomes and metagenomes. We now explicitly recommend that users and potential tool developers employ at least a minimal low-complexity filtering. From a benchmark perspective, this helps isolate tool-specific complexity handling (not all tools allow manually disabling such settings). From a practical perspective, we identified that certain exhaustive tools (or tools with im-

mutable cost matrices) create massive outputs with some low-complexity sequences, for instance when aligning sequences containing ambiguous "N" bases.

In the current version of the "real-data" analysis, we employ very light complexity filtering: no ambiguous bases, no spacers where a single base frequency exceeds 90% of the sequence (e.g., homopolymers such as polyA/T/G/C), and no spacers with highly repetitive subsequences.

6. **Stratified subsampling approach:** Partially due to computational constraints and to test tool behavior at different scales, we modified the "real" dataset analysis (IMG/VR4 contigs vs iPHoP spacers). Instead of using the full contig set, we employed a stratified subsampling strategy with fractions of varying sizes (0.001, 0.005, 0.01, 0.05, 0.1 and 1 - albeit not all tools successfully completed the analysis for all fractions in reasonable time) that maintains the original distribution of contigs' taxonomic class labels. We observe consistent trends within different subsampled fractions for recall and resource usage, although notable exceptions were strobealign, mmseqs, and x-mapper. However, even in subsamples where these performed relatively well (compared to themselves), they are not currently compelling tools for this task. We therefore avoided investigating their inconsistencies as it is outside the scope of this project. The same consistency for the majority of the tools was observed in the different fully simulated datasets (see Supplementary Figures).

7. **Enhanced synthetic dataset:** As suggested by reviewers, we modified the synthetic dataset generation to include more realistic parameters, including contig length and GC distributions matched to IMG/VR4 viral contig set and the Iphop spacer set. We include an additional jupyter notebook in the supplementary material that demonstrates the sequence characteristics (nucleobase frequencies, lengths, types of distributions, k-mer patterns, entropy and complexity) between the real and synthetic spacer datasets. Briefly, the real and simulated sequences appear to have similar feature distributions, although the real spacer set tends to include some low complexity sequences (albeit these are relatively rare, and we now attempt to minimize their effect on the analyses).

We also added a new "semi-synthetic" simulation run, where the presence of real spacers in fully synthetic contigs is tested. We use this to estimate the rate of non-specific matches (previously termed "spurious" matches) in a more realistic setting.

8. **Hamming vs Edit Distance Analysis:** We corrected and clarified the terminology around distance metrics, true-positive sets, and invalid sets. Specifically, the previously termed "spurious alignments" in the synthetic set (matches occurring in regions where the simulation did not explicitly insert a spacer) are now used as a proxy for estimating the rate of matches arising by chance, given a distance threshold and metric (hamming or edit).

This is conceptually similar to an e-value: 'given a query sequence and a target sequence set, what is the expected number of matches with <= k distance by chance?'

We conducted a comprehensive analysis comparing hamming distance (substitutions only) versus edit distance (allowing indels), and reviewed existing literature on phage "escape" mutations of protospacers (added to the discussion and introduction). For edit distance measurements, we used sassy - the only tool supporting arbitrary edit distances with perfect recall (exhaustive), and the aggregate of the heuristic tools (namely bowtie1 and blastn, previously demonstrated to have good recall) for hamming distance measurement of very large sequence sets as a proxy (the exhaustive tools could not complete within reasonable time).

Using both synthetic and semi-synthetic datasets, we estimated the frequency of valid-but-unplanned matches (i.e., alignments verified to be within a given distance threshold, regardless of which tool predicted them, but not explicitly inserted in the simulated contigs). Based on the simulated runs, we observe that such unplanned matches are typically an order of magnitude more frequent when searching up to n edit distance compared to n hamming distance, and this trend increases considerably above distance 3 (see Supplementary Figure S9, Supplementary Tables S8 and S7).

While this is expected (all hamming alignments are a subset of all edit-distance alignments), the exact rate and difference in the number of permitted alignments under these metrics is difficult to estimate. For practical purposes, a value is often approximated for the frequency of matches arising by chance (e.g., e-value). Our use of empirical data (the simulation runs) is partially justified by the new simulation's ability to mimic real biological sequences more closely (with regard to base distribution, lengths, etc.). For example, using the semi-synthetic dataset (measuring the occurrence of ~3.8M real spacers in ~421k simulated contigs, matching the IMG/VR4 set size), we identified 56,273 validated non-planned alignments at hamming distance <=3, 2,354 at <=2, 49 at <=1, and 1 for exact matches.

9. **Biological justification for hamming distance:** We added a brief literature review and argument for preferring hamming distance in most scenarios. Most experimental phage-host studies report that escape mutations are predominantly single substitutions, in the protospacer or in the PAM-proximal "seed" region. While some indels have been reported, we could not find quantitative comparisons across mutation types; we identified only one study mentioning indel escape mutations in protospacers (Paez-Espino et al., 2015), and to the best of our understanding the reported case involves a single indel mutation without comparative quantification. From a structural perspective, phage genomes are coding-dense, and a single indel in a coding sequence causes a frameshift mutation that is often lethal, whereas a substitution may only affect a single amino acid residue. Consistent with this, the literature suggests indels are approximately 10-

fold rarer than substitutions in bacteria, and we expect similar ratios in phages. We also distinguish biological indels from sequencing artifacts. With sufficient sequencing depth, sequencing-induced indels should be rare in assembled contigs, particularly for Illumina-based datasets where consensus sequences are used. However, for other sequencing technologies (e.g., Oxford Nanopore) or low-depth datasets where raw reads are analyzed directly, some tolerance for indels may be necessary. We note that while long-read technologies are attractive for CRISPR arrays (which are repetitive and prone to misassembly in short reads), the alignment accuracy cannot exceed the underlying sequencing and base-calling accuracy. We further acknowledge that most existing studies focus on substitutions, which may create an assumption bias where indels are underexplored; future experimental work could systematically compare escape mutation types. Another reported escape mechanism involves large genomic rearrangements (e.g., gene order changes at spacer-targeted boundaries) (Paez-Espino et al., 2015; Kupczok et al., 2018), but these would not be detectable with a small edit distance threshold given the substantial sequence rearrangements involved. We explicitly address cases where edit distance should be considered, such as when using low-accuracy long-read data or when investigating escape mutation mechanisms under verifiable experimental settings.

Response to Reviewer 1

Major Concerns

1. Enhanced Synthetic Dataset Analysis and Realistic Scenarios

"the study felt like a missed opportunity to study how realistic scenarios and biased datasets affect the detection of spacer-protospacer pairs… the statistical properties of the synthetic dataset are very simplistic and far from representing real databases."

Response: We enhanced our synthetic dataset generation to address these concerns. We now generate contigs with realistic length distributions based on IMG/VR4 data rather than uniform distributions, having added to the CLI tool the option to simulate lengths under uniform or normal distributions. In the simulated datasets used in the revised version, we roughly matched these distributions to the real IMG/VR4 contig length range. Similarly, we added options for controlling the base composition, and matched the GC% to that of the iPHoP spacer data (~49% for spacers, ~46% for contigs). We ran additional comparisons to show that other sequence characteristics (k-mer distributions, repeat abundance, etc.) are also similar between the real and the new synthetic datasets, albeit the simulated sequences still appear to have a narrower distribution for certain attributes (for example, number of repeated k-mers). We acknowledge this shortcoming in the study limitation section, and added distribution plots of different sequence characteristics in the supplementary data (Supplementary Figures S5 and S4).

2. False Positive Analysis and Precision Metrics "...I missed a more honest treatment of false positives. If you make a synthetic dataset, select pre-planned insertion points and then find out that the dataset, which is completely random (no sequence conservation, shared ancestry, or HGT), contains other identical or nearly identical sequences, then those can only be considered false positives. The authors even call these"spurious alignments", which is quite revealing of what they actually are (or what they are not: meaningful or informative). The fact that the number of those spurious alignments skyrockets with the mismatch threshold is also quite revealing: the greater the number of accepted mismatches, the greater the risk of getting "protospacers" that are not meaningful. The fraction of such hits in the synthetic dataset is quite remarkable (>10%) if one allows up to 3 mismatches. The fraction in a more realistic (synthetic) dataset with compositional bias would be even higher (because biases reduce the total entropy). From a biological point of view, such meaningless hits are false positives and should be treated as such. Thus, I was puzzled that the authors not only did not leverage the synthetic dataset to assess the specificity of different tools against false positives (which perhaps could make blastn-short perform better than bowtie1) but also counted them as true positives. I agree that doing that with the real dataset is complicated, because deciding whether a hit is meaningful or spurious would require further analyses of phylogeny and genomic context. But not doing that with the synthetic dataset seems harder to justify. In the end, finding spacer-protospacer pairs with mismatches comes at the cost of getting more false positives. Therefore, depending on the application, better performance finding pairs with 2-3 mismatches may not be that advantageous."

Response: We agree with the reviewer and have revised our analysis. As mentioned above, we have clarified the terminology and metrics used. We now use these "spurious alignments" (after validating that they are not erroneous tool reports) as a proxy for the rate of non-informative matches with regard to biological origin (such as would be expected for true host-phage interactions). As the reviewer notes, the ">10%" rate is indeed quite remarkable. However, this percentage estimates the rate of non-planned matches relative to planned spacer occurrences. Since we control the number of planned occurrences, this percentage is not the most useful metric. We now instead report either the absolute number of such cases or their frequency normalized by the target (contigs) and query (spacers) set sizes. We perform this for both edit and hamming distances, although as noted in the text, for the semi-synthetic datasets, none of the exhaustive tools could finish within a generous allocated time.
Specifically, using the semi-synthetic dataset (~3.8 million real spacers as queries, searched against ~421k simulated contigs matching IMG/VR4 sequence characteristics), we identified: 56,273 unique alignments at hamming distance <=3, 2,354 unique alignments at <=2, 49 alignments at <=1, and 1 exact match (distance 0). We observe similar rates, albeit slightly lower rates (when adjusted for sequence length) in the fully synthetic runs (more details in the Results section "Selection of distance metric and threshold values").

3. Real Dataset Limitations and Positive Set Definition

"The positive set for the real dataset is just the union of the protospacers identified by any of the tools. Thus, protospacers that are missed by all tools are completely ignored when estimating the recall."

Response: We agree that this is an important point, and now note this limitation explicitly in the text and explain that the majority of the tools tested are not exhaustive (i.e., they use heuristics in the sense it is not guaranteed they will report all matches). We have specifically added subsample analyses small enough to include exhaustive tools, and their results are part of the union of all results. For larger subsets, these tools were unable to finish even after 72h x 64 CPUs. Overall, the relative recall rates between tools appear consistent with and without adding the union of the exhaustive tool results. That said, we now explicitly mention that without the exhaustive tool results, we are likely underestimating the actual number of valid alignments.

4. Tool Design Features and Performance Insights

The reviewer asked for "insight that connected the heuristics and design features of different methods with their performance" and suggests "adding more information to Table 1."

Response: We have added information regarding intended use and tool overview in several sections of the manuscript. Table 1 now includes tool categorization (exhaustive vs heuristic search), original intended purpose, and the notable configuration used in our analysis. In the introduction, we include an overview of the computational background for sequence similarity, discussing the main algorithms used by the different tools and their biological relevance (i.e., use cases). We also note how certain tool-specific heuristics can affect performance under different definitions, which are often goal-oriented (for example, tools built for identifying the best match under the assumption of a single reference source). In this regard, we note that our specific use of these tools is not their original intended purpose.

Minor Points

5. Methodological Clarifications

Regarding distance metric terminology and biological justification, we clarify that the parasail-based re-alignment and distance count method differed from true Levenshtein distance. In the new version, we utilize the edlib library for finding the best alignment that includes gaps and measure the edit distance for it, and we compare that to hamming distance now measured as the sum of substitutions and clipped/unaligned spacer bases from the best ungapped alignment. Additionally, we added explicit options in the CLI tool for re-alignment and distance measurement with a third option: gap-affine with custom cost matrix, gap open, and extension values (for alignment), but in the current version, we did not use it and focus instead on the minimal edit distance (using edlib), and the simplified, correct hamming.

Regarding minimap2 performance, the reviewer correctly notes that minimap2 displays near-zero recall. We adjusted the parameters as suggested (lowered `-m` values, specifically tried `-k 8`, `-w 6`, `-P -m 10`) and indeed obtained more matches compared to the previous arguments. However, minimap2 still underperformed considerably, displaying near-zero recall compared to most other tools. We therefore decided to omit the minimap2 results from the main text, as this appears to be an artifact of un-optimised parameter selection, and including these near-zero results would be misleading and uninformative. That being said, we generated a version of all the supplementary and main figures with minimap2 results included, for reviewer information. We note that while minimap2 is apparently not well-suited for CRISPR spacer-protospacer matching in its current configuration, it remains an excellent tool for its intended purposes and is very popular for read mapping and similar tasks (hence its inclusion in the benchmark in the first place). We also note that we modified parameter choices for several other tools, and we clarify in the manuscript that we selected parameters to maximize recall for short (spacer-length) queries, based on our understanding of the tools and their guidelines (documentation, GitHub issue threads, etc.). We added a note in the study limitations that it is possible that exploring all possible tool-parameter combinations might yield different results. We hope that in future versions of this project, tool developers will directly inform what parameters they recommend.

Concerning high target multiplicity and database redundancy, the reviewer raises an important point about whether high-multiplicity spacers (>1000 targets) are informative, noting that "…spacers with >1000 targets are probably not very informative (unless the MGE database is extremely redundant)". While these are rare (Supplementary Figure S6), we clarify that both the contig and spacer datasets used (in the original analysis and in this revision) are deduplicated (non-redundant) at the sequence level (we explicitly only include unique sequences, removing duplicates or reverse-complement sequences). The high multiplicity reflects genuinely shared short sequences appearing across different contigs, not sequence database redundancy from identical full-length contigs. This phenomenon is biologically meaningful and has been documented in the literature. For example, "frozen prophages" – prophages appearing in certain bacterial lineages that are targeted by large CRISPR arrays from the same bacteria – create legitimate high-multiplicity scenarios, as these can align well with related prophages. We discuss this in the manuscript (referencing Stern et al. and others), noting that such cases represent real biological patterns rather than database artifacts. However, we acknowledge that interpretation of very high-multiplicity spacers requires careful consideration (as they may represent conserved genomic regions rather than specific phage-host interactions). Furthermore, we have estimated that these high-occurrence spacers are too rare (less than 0.1% of all spacers, Supplementary Figure S6) to dictate tool choice. In the revised manuscript, we focus more on the distance value effect on recall rate and the feasibility of selecting exhaustive vs heuristic tools, and we dedicate a considerably smaller section to variations caused by such spacers.

6. Figure and Presentation Issues

Response: We have addressed the figure and presentation concerns, specifically: we now use standardized tool-specific colors and symbols across all figures in both the main text and supplementary material. For improved visibility, the distance-versus-recall comparison is now presented as a heatmap (Figure 2), replacing the previous scatter and line plots where overlapping lines made individual tools difficult to distinguish. Pairwise tool-vs-tool comparison matrices have been moved to the supplementary material (Supplementary Figure S1). The Figure 2 caption has been revised to remove the circular definition of "detection fraction". The Methods section has been reorganized with dedicated subsections, and all figure references and numbering have been corrected throughout the manuscript and supplementary material.

Response to Reviewer 2

Major Points

1. Title and Computational Focus

The reviewer suggests the title "should reflect computational nature of study and purposes of detection" and use "a more appropriate word than drastic."

Response: We have revised the title to "Computational Tool Choice Impacts CRISPR Spacer-Protospacer Detection".

2. Introduction: Prior Work and Citations

The reviewer notes: "Most of these points have been examined using bioinformatic and experimental tools before and these should be cited. Reference and comment on the basis of current tools as well as older versions and reviews."

Response: We have substantially expanded the introduction to cite and discuss prior work on CRISPR spacer-protospacer matching, the biological mechanisms of phage escape mutations (and notable experimental work in this field) , and the computational foundations of sequence similarity searches. In particular, we now cite foundational experimental studies on escape mutations (Deveau et al. 2008, Semenova et al. 2011, Fineran et al. 2014, Schelling et al. 2023, Paez-Espino et al. 2015), mutation rate studies in phages and in their hosts (Lee et al. 2012, Kucukyildirim et al. 2021), phage genome structure references (Hatfull & Hendrix 2011, Ha et al. 2018, Kupczok et al. 2018), and CRISPR-related host-prediction tools and databases (Edwards et al. 2015, Biswas et al. 2013, Roux et al. 2023, Zhang et al. 2021). We also cite algorithmic references for the evaluated tools and the classical sequence alignment literature (Needleman-Wunsch 1970, Smith-Waterman 1981, Myers 1999). The introduction now includes dedicated subsections on computational foundations, heuristic algorithm design, distance metrics, and false positive considerations.

3. DUST Citation and Low-Complexity Filtering

The reviewer asks: "Cite DUST if it has been used in this type of application before. Shannon entropy is not important here, base pairing is."

Response: We now cite DUST (Morgulis et al. 2006) and tantan (Frith 2010) in both the introduction and discussion as relevant complexity filtering tools. Note that in the revised manuscript, we performed minor pre-filtering (removing <2% of the spacers) on the real-data spacer set. Our rationale for this is to provide a uniform starting point across tools, as some tools handle complexity differently (some have internal hard-coded methods to mask these, some tools do not perform any such filtering, and some tools enable turning these off via user supplied CLI argument). We clarify that in the revised work, Shannon entropy is used as one of several metrics in our pre-filtering step to identify very low-complexity spacers, alongside extreme base frequency thresholds (i.e. near complete homo-polymer) and highly repeated multimers. We note that these are proxies, rather than a primary analysis metric or goal.

4. BLAST 2018 Issues

The reviewer notes: "The blast issues reported in 2018 appear to be resolved with later bug fixes and better documentation. They need not be reiterated here, unless discussing specific current issues or parameters."

Response: We have revised the relevant paragraph in the introduction to explicitly note that the bugs reported by Shah et al. (2018) were subsequently patched, and that Madden et al. (2018) clarified the situation as a combination of a since-fixed software bug and misconceptions about BLAST+ tie-breaking. We retain a brief mention because the episode illustrates a broader point relevant to our work: assumptions about the exhaustiveness of tool result/reporting can lead to incorrect conclusions, and this is directly pertinent to our benchmark where reporting completeness is our key performance metric.

5. Enhanced BLASTn Analysis and Parameter Testing

The reviewer emphasizes that "blastn results are most relevant to many current tools and analyses" and requests "more analysis of blastn's performance" ... "I note that some spacer matching tools e.g. CRISPRTarget use unusual blastn parameters for matching, albeit for a different purpose and on a smaller scale (https://crispr.otago.ac.nz/CRISPRTarget/crispr_analysis.html) some other sets of parameters should also be tested e.g . wordsize if not as small as possible (7) or -evalue."

Response: We clarify that the parameters used for BLASTn in our analysis were specifically chosen to enable the highest possible recall for short spacers - specifically blastn-short (-task blastn-short, sets word-size at 7, compared to 11 with default blastn, and 28 in megablastn) and a very high number of max target sequences (-max_target_seqs 1000000, so up to 1M reported alignments per query, compared to the default value of 500). After some internal experimentation, we have updated the parameters when calling blastn, adding the '–ungapped' flag, and we set perc_identity=84 and qcov_hsp_perc=80. The ungapped flag is used by CRISPRTarget, aligns with our current recommendation of hamming (and not edit) distance, and we anecdotally observed reduced total cpu time with this option. The perc_identity=84 and qcov_hsp_perc=80 arguments are mostly used to control the output size, as we do not know of a way to restrict blastn to a specific "n" distance. These values should allow for a range of distances to be reported that exceeds our desired threshold (<=3) in order to capture all valid matches. We add a paragraph in the methods section clarifying this choice compared to the default parameters and parameters used by certain tools and previous studies (e.g. the original IMG/VR4 spacer-protospacer mapping used default blastn with -max_target_seqs 1000). Regarding "CRISPRTarget" - the legacy webserver is not available anymore (per https://crispr.otago.ac.nz/CRISPRTarget/crispr_analysis.html).The research paper describing CRISPRTarget (Biswas et. al. 2013) mentions that the website allowed users to select custom values for word size, evalue and choice of target database for the BLASTn step, but recommended using the default blastn-short values for word size , with a permissive e-value (to account for variations in database size) and modified gap penalties: "The initial CRISPRTarget defaults are the same except that a gap is penalized more highly (-10), the mismatch penalty is -1 and the E filter is 1. In addition, there is also no filter or masking for low complexity". These adjustments should favour gapless alignments compared to blastn-short defaults (gap open -5, gap extend -2). While we do not disagree with these parameter choices, we prefer to use the qcov and %id options to control the reported alignments. We added a paragraph addressing this in the main text.

7. DUST Filtering / Low Complexity Analysis

The reviewer suggests testing "the effect of DUST" and asks "are the sequences with high abundance being missed of low complexity."

Response: We have implemented a light pre-filtering step (spacer_inspection.ipynb) using several metrics to pre-emptively measure and discard low-complexity, repetitive, or homopolymer spacers prior to any mapping/alignment/benchmarking. We added a paragraph regarding this to the methods section and now explicitly note this as a key recommendation for future analyses.

We further note that while some tools employ internal complexity filtering (e.g., DUST in blastn, tantan in mmseqs2), not all tools do so consistently or allow us to disable these filters. Therefore, applying pre-filtering uniformly across all tools ensures comparable input handling.

Regarding whether the high-abundance spacers being missed are low-complexity: we note that high-abundance spacers (>1000 occurrences) are still observed in the reanalysis of this version, indicating they pass the complexity filters.

8. Tool Recommendations and Guidelines

The reviewer notes that "the authors state they provide general guidelines in the abstract but these are less clear in the discussion."

Response: We have substantially expanded the discussion and conclusion sections to provide explicit, scale-dependent tool recommendations. For most large-scale metagenomic analyses where the total contig set exceeds approximately 1 Gbp and the spacer set may contain millions of sequences, we recommend Bowtie1 at hamming distance <=3, given its combination of high recall among heuristic tools, favorable computational efficiency, and consistency across dataset scales (Figure 2, Figure 3). For analyses requiring tolerance beyond 3 substitutions, indelfree.sh in indexed mode extends the hamming distance range (up to 5) without the elevated non-planned match rates associated with edit distance. For targeted analyses of specific host-phage systems, lineage-specific comparisons, or any setting where the total subject sequence is below approximately 1 Gbp, exhaustive tools such as Sassy and indelfree.sh bruteforce remain tractable and provide the advantage of guaranteed complete detection; they can also serve to validate heuristic tool outputs when computational resources permit. For large-scale analyses using complete public spacer and virus databases, a combination of heuristic tools currently appears as the only computationally feasible option.

We restrict our recommendation for edit distance metrics to narrowly defined conditions: when analyzing data from sequencing platforms with elevated indel error rates (e.g., Oxford Nanopore R9 chemistry) where sequencing-induced indels cannot be excluded, when explicitly characterizing escape mutation mechanisms in controlled experimental settings where the mutation type itself is informative, or in gene-editing off-target prediction contexts where gapped alignments have established biological relevance. Outside these scenarios, the substantially higher non-planned match rates associated with edit distance (6-8 fold at <=3) argue against its routine use for inferring host-MGE interactions from natural populations. We also discuss computational resource trade-offs (Figure 3, Supplementary Figure S7), noting that Bowtie1 and indelfree.sh in indexed mode scale well to large databases while exhaustive tools are limited to smaller datasets. Finally, we provide specific parameter recommendations for BLASTn, where the choice of -max_target_seqs can substantially impact high-abundance spacer detection.

Minor Points

6. Figure Improvements

Response: We have addressed all figure-related concerns. Tool-specific symbol styles and colors have been improved for visual distinction. We corrected typos and improved legends in the supplementary figures, and ensured that colors and symbols representing each tool are consistent across all figures in the main text and supplementary material.

Additional: Historical Comparison

The reviewer notes: "which would 'allow for a direct comparison to historical results'. Was that comparison ever made? I could not find any further mention to that in the manuscript."

Response: Our original choice to benchmark tools using the IMG/VR4 contig dataset is inherently a comparison with the original IMG/VR4 publication, which used spacer-protospacer matching for host assignment (using BLASTn with -max_target_seqs 1000). In the current version, we removed the explicit mention of comparison to historical results, as we are now using a different spacer sequence set (iPhoP June 2025).

9. Reference Genome Spacers and Known Prokaryotic Viruses

The reviewer notes: "This study focuses only on metagenomic data and is useful as such. It would be useful to see a set of spacers from reference genomes, and known prokaryotic viruses. Some of the effects seen here could be due to biases in the spacer library or IMGVR datasets."

Response: The datasets used in the revised manuscript include sequences from both reference genomes and metagenomic sources. The iPHoP spacer set (June 2025 release) explicitly combines CRISPR spacers extracted from both reference genomes and metagenomes, as described in the Methods. Similarly, IMG/VR4 viral contigs are supplemented with sequences from NCBI RefSeq and GenBank, which include known prokaryotic viruses with cultured isolate genomes. Thus, the benchmark is not exclusively metagenomic in origin.

Additionally, the simulated datasets provide a complementary evaluation under controlled sequence composition (with known ground truth), which is independent of any biases present in the real spacer or contig databases. The consistency of tool behavior across synthetic, real, and semi-synthetic datasets (as shown in Figures 1 and 2, and the supplementary figures) suggests that the observed performance differences reflect genuine algorithmic properties rather than dataset-specific artifacts. We acknowledge that a dedicated analysis focusing exclusively on reference-genome-derived spacers matched against characterized phage isolates would be a valuable complementary study, but consider it beyond the scope of the present work, which is focused on benchmarking tool performance under conditions representative of current large-scale metagenomic analyses.

Supplementary Material

All new notebooks, datasets, and analysis scripts are provided in the public github repository (https://github.com/UriNeri/spacer_matching_bench). The final mapping results (i.e. files too large for github) are uploaded as a new version to the same Zenodo deposit as the previous version (DOI: 10.5281/zenodo.15171878)

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